

ASSIGNMENT NO.9: Data Retrieval, Functions, and Joins in MySQL

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: OnlineEnrollment and OfflineEnrollment.

OnlineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

OfflineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

Insert at least 6 records into each table. Some records should be common in both tables.

Query-

```
CREATE TABLE OnlineEnrollment (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    CourseID VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT  
);
```

```
CREATE TABLE OfflineEnrollment (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    CourseID VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT  
);
```

```
INSERT INTO OnlineEnrollment  
(StudentID, StudentName, CourseID, City, Marks)  
VALUES  
(1, 'Rahul', 'DS101', 'Pune', 85),
```

```
(2, 'Priya', 'CS102', 'Mumbai', 90),  
(3, 'Amit', 'IT103', 'Nashik', 72),  
(4, 'Neha', 'AI104', 'Pune', 88),  
(5, 'Riya', 'DS101', 'Mumbai', 76),  
(6, 'Karan', 'CS102', 'Pune', 82);
```

```
INSERT INTO OfflineEnrollment  
(StudentID, StudentName, CourseID, City, Marks)  
VALUES  
(1, 'Rahul', 'DS101', 'Pune', 85),  
(2, 'Priya', 'CS102', 'Mumbai', 90),  
(4, 'Neha', 'AI104', 'Pune', 88),  
(7, 'Aarav', 'IT103', 'Delhi', 79),  
(8, 'Sneha', 'DS101', 'Nagpur', 84),  
(9, 'Vikas', 'CS102', 'Mumbai', 81);
```

2) Write a MySQL query using UNION to display all unique students from both OnlineEnrollment and OfflineEnrollment.

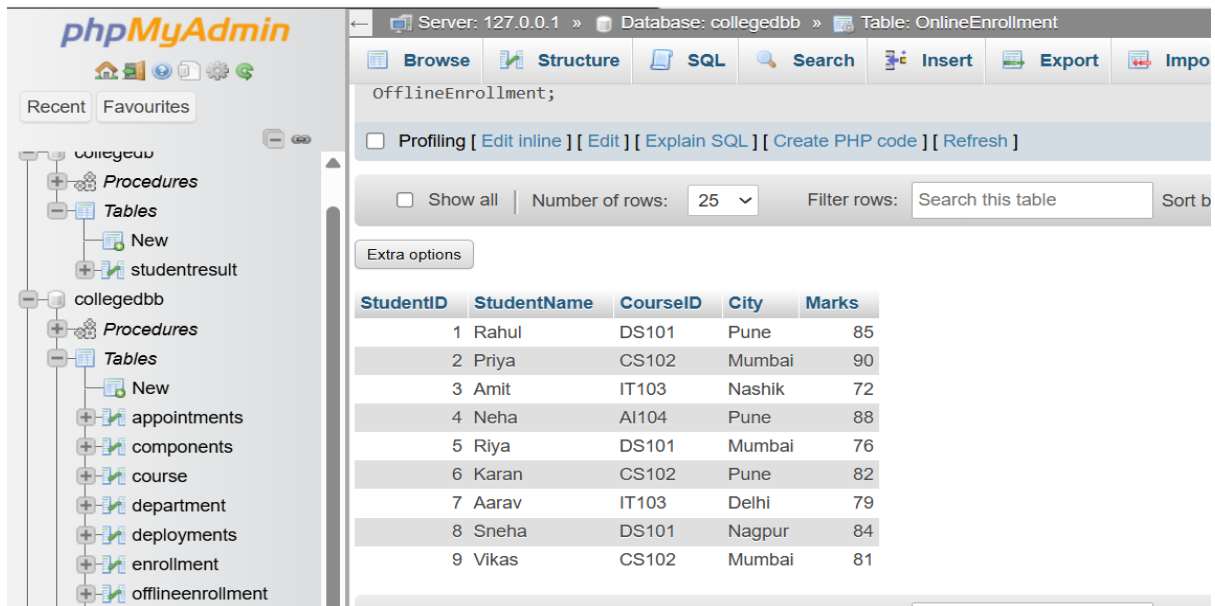
The output should contain:

Column
StudentID
StudentName
Course
City
Marks

Query-

```
SELECT StudentID, StudentName, CourseID, City, Marks  
FROM OnlineEnrollment  
UNION  
SELECT StudentID, StudentName, CourseID, City, Marks  
FROM OfflineEnrollment;
```

Output-



Server: 127.0.0.1 » Database: collegedbb » Table: OnlineEnrollment

OfflineEnrollment;

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☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort b

Extra options

StudentID	StudentName	CourseID	City	Marks
1	Rahul	DS101	Pune	85
2	Priya	CS102	Mumbai	90
3	Amit	IT103	Nashik	72
4	Neha	AI104	Pune	88
5	Riya	DS101	Mumbai	76
6	Karan	CS102	Pune	82
7	Aarav	IT103	Delhi	79
8	Sneha	DS101	Nagpur	84
9	Vikas	CS102	Mumbai	81

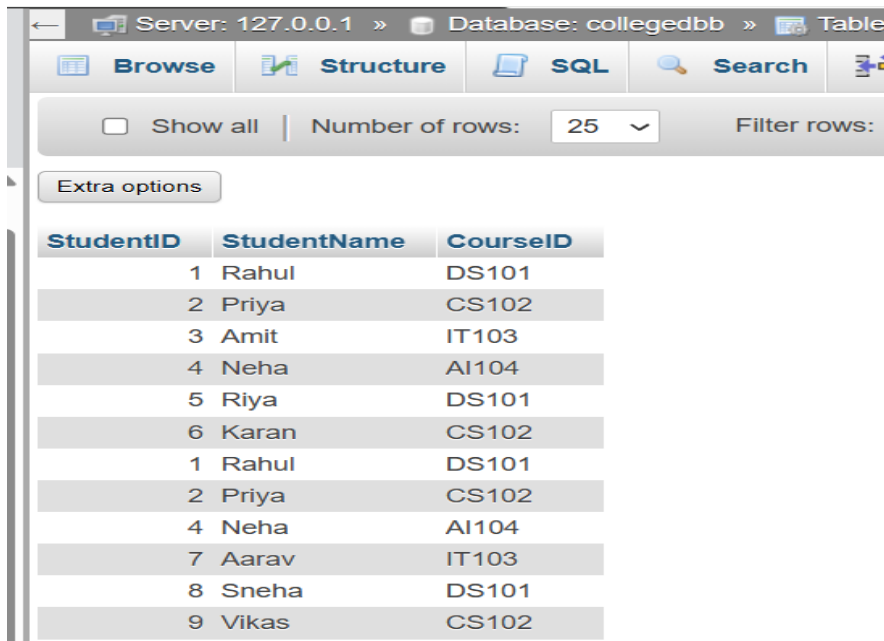
Q3) Write a MySQL query using UNION ALL to combine records from both tables while keeping duplicate records.

Also write one line explaining the difference between UNION and UNION ALL.

Query-

```
SELECT Student ID, StudentName, Course
FROM OnlineEnrollment
UNION ALL
SELECT StudentID, StudentName, Course
FROM OfflineEnrollment;
```

Output-



Server: 127.0.0.1 » Database: collegedbb » Table:

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

StudentID	StudentName	CourseID
1	Rahul	DS101
2	Priya	CS102
3	Amit	IT103
4	Neha	AI104
5	Riya	DS101
6	Karan	CS102
1	Rahul	DS101
2	Priya	CS102
4	Neha	AI104
7	Aarav	IT103
8	Sneha	DS101
9	Vikas	CS102

UNION removes duplicate records, whereas **UNION ALL** keeps duplicate records.

Q4) Write a MySQL query using INTERSECT to display students who are present in both online and offline enrollment tables.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

Query-

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OnlineEnrollment
INTERSECT
SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```

Output-

✓ Showing rows 0 - 2 (3 total, Query took 0.0019 seconds.)					
<pre>SELECT StudentID, StudentName, CourseID, City, Marks FROM OnlineEnrollment INTERSECT SELECT StudentID, StudentName, CourseID, City, Marks FROM OfflineEnrollment;</pre>					
☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]					
☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None					
Extra options					
StudentID	StudentName	CourseID	City	Marks	
1	Rahul	DS101	Pune	85	
2	Priya	CS102	Mumbai	90	
4	Neha	AI104	Pune	88	
☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None					

Q5) Write a MySQL query using EXCEPT to display students who are present only in the online enrollment table.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

Query-

```
SELECT StudentID, StudentName, CourseID, City, Marks
FROM OnlineEnrollment
EXCEPT
SELECT StudentID, StudentName, CourseID, City, Marks
FROM OfflineEnrollment;
```

Output-

✓ Showing rows 0 - 2 (3 total, Query took 0.0017 seconds.)

```
SELECT StudentID, StudentName, CourseID, City, Marks FROM OnlineEnrollment EXCEPT SELECT StudentID, StudentName, CourseID, City, Marks FROM OfflineEnrollment;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID	StudentName	CourseID	City	Marks
3	Amit	IT103	Nashik	72
5	Riya	DS101	Mumbai	76
6	Karan	CS102	Pune	82

Q6) Write a MySQL query to display high-scoring students from both tables.
The query should:

Requirement
Select students with marks greater than or equal to 85
Use UNION to combine both tables
Display StudentID, StudentName, Course, and Marks
Sort the final output by Marks in descending order

Query-

```
SELECT StudentID, StudentName, CourseID, Marks  
FROM OnlineEnrollment  
WHERE Marks >= 85  
UNION  
SELECT StudentID, StudentName, CourseID, Marks  
FROM OfflineEnrollment  
WHERE Marks >= 85  
ORDER BY Marks DESC;
```

Output-

✓ Showing rows 0 - 2 (3 total, Query took 0.0024 seconds.)

```
SELECT StudentID, StudentName, CourseID, Marks FROM OnlineEnrollment WHERE Marks >= 85 UNION SELECT StudentID, StudentName, CourseID, Marks FROM OfflineEnrollment WHERE Marks >= 85 ORDER BY Marks DESC;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID	StudentName	CourseID	Marks	1
2	Priya	CS102	90	
4	Neha	AI104	88	
1	Rahul	DS101	85	